CLAIMPROMOTEBENCH: COMPILING EXECUTABLE POLICY REVISIONS INTO PAIRED MODEL-SERVICE REGRESSION TESTS

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ABSTRACT

Policy revisions can change prescribed actions for fixed evidence, creating regression risk in structured decisions routed through language-model services. CLAIMPROMOTEBENCH is a synthetic, policy-constrained evaluation that compiles two executable policy versions and a finite grammar into paired tests that hold context fixed and change only policy. It certifies changed-rule and action-transition coverage and selects a minimum joint suite. Two source-traceable Gatekeeper revisions yield 12 pairs covering two rules and both transitions; two pairs suffice. Across three frozen repetitions per hosted route, strict end-to-end action/pair scores are 53/72 and 21/36 for DeepSeek Flash and 10/72 and 0/36 for Qwen. Every extracted action is correct; failures are 12 timeouts and 69 output-envelope violations. A 1,320-item audit recovers 49 correct actions and 40 correct pairs outside the full contract. R6 Qwen scores 258/300 items and 113/150 pairs end to end versus 258/272 and 113/123 conditional. On a public balanced challenge, Qwen and Flash score 132/180, 48/90 and 90/180, 13/90 end to end versus 132/148, 48/64 and 90/146, 13/61 conditional; `DEMOTE` recall is 4/30 and 0/30. Results support regression testing of finite authored policy surfaces and model-service interfaces.

1 INTRODUCTION

Deployed decision policies evolve. Content-moderation studies describe frequent policy updates, limited new labels, and conflicts between historical labels and current rules (Mullick et al., 2023; Ji et al., 2026). Software-policy repositories expose the same maintenance problem in executable form. A Gatekeeper revision changed an empty FlexVolume allowlist from permitting every driver to blocking every driver; another restricted a selector search to Services (Open Policy Agent Gatekeeper Library, 2020; 2021). Trivy corrected an inverted admission predicate (Aqua Security Trivy Checks, 2024), and Cosign repaired a verifier that could accept a true `deny` result or an undefined query (Sigstore Cosign, 2022). These cases establish that policy and verifier revisions can change operational actions for fixed inputs. Their prevalence remains unmeasured.

Some applications place a language-model service between structured evidence and a downstream action. This boundary appears during migration, in legacy integrations, or when one route handles both natural-language and structured inputs. An executable policy should remain the production decision mechanism whenever the application can call it directly. For model-mediated boundaries, the policy executor can instead serve as a regression oracle: after a policy update, does the route follow the new decision surface, and does it still satisfy the downstream output contract?

Static accuracy gives an incomplete answer. A route can ignore the policy version and exploit stable field–action correlations. A response can also expose the correct action while violating a larger schema, or fail before any action is available. Valid-only scoring removes those interface failures from the denominator. Rule coverage creates another blind spot: a suite may exercise every edited rule while omitting action transitions induced by their interactions.

CLAIMPROMOTEBENCH converts a policy revision into paired regression tests. Given executable policies P_0, P_1 and a finite context grammar $\mathcal{X}$, the compiler enumerates contexts where the prescribed action changes. It duplicates each context and varies only the active policy. A pair is correct only when both actions follow their respective policies. The compiler reports two oracle-

side certificates—changed-rule coverage and directed action-transition coverage—and selects a cardinality-minimum suite that meets both. Route evaluation then separates safe action extraction, full contract validity, and exact structured-decision agreement while retaining every scheduled call in end-to-end denominators.

The evaluation has three evidence layers. First, exact upstream source revisions from Gatekeeper, Trivy, and Cosign establish external fault instances. Second, the Gatekeeper revisions supply a frozen 12-pair model-service study with three repetitions on two hosted routes. Third, an authored four-action SLSA-inspired migration stress-tests transition coverage, and two research-evidence suites diagnose action imbalance and endpoint sensitivity. The latter policies and labels are author-defined instruments.

The contributions are:

1. a compiler that turns an executable policy revision into paired model-service tests with changed-rule and action-transition adequacy certificates;
2. a failure-inclusive evaluation protocol with distinct strict-action, contract-action, and full-decision endpoints at item and pair level; and
3. source-traceable validation on two real Gatekeeper revisions, independent Trivy and Cosign fault reproductions, and repeated route measurements whose schedules and terminal outcomes are frozen and auditable.

2 Related Work

Behavioral tests and contrasts. CheckList organizes behavioral tests by capability, and contrast sets probe local decision boundaries through controlled input changes (Ribeiro et al., 2020; Gardner et al., 2020). Metamorphic testing evaluates relations among transformed executions (Chan et al., 1998). ClaimPromoteBench uses a policy-version intervention and an exact executable oracle; its paired score detects routes whose action remains invariant across an action-changing revision.

Policy differencing and regression tests. Margrave computes witnesses for access-control policy disagreement, and Cirg applies change-impact analysis to generate and reduce policy regression tests (Fisler et al., 2005; Martin & Xie, 2007b). Mutation testing and suite reduction provide established adequacy machinery (Martin & Xie, 2007a; Harrold et al., 1993). These works establish the semantic-difference construction used here. ClaimPromoteBench applies that construction to a model-service boundary and adds paired route records, action-transition adequacy, structured-output endpoints, and complete service-failure denominators.

Evolving rules and agent specifications. GuideBench evaluates domain guidelines, including rule updates (Diao et al., 2025). TDAD compiles behavioral specifications into agent tests and studies specification evolution with hidden tests and semantic mutants (Rehan, 2026). Policy-to-Tests translates governance prose into executable rules (Datla et al., 2025). ClaimPromoteBench starts from two deterministic policy versions. It computes their behavioral difference and preserves the policy version as the controlled coordinate in each pair. This scope isolates regression testing from policy extraction, prompt optimization, and judge-based labeling.

Structured output and selective prediction. Grammar-constrained decoding and incremental parsing improve syntactic validity (Scholak et al., 2021; Geng et al., 2023); JSONSchemaBench and BFCL evaluate schemas and tool arguments (Geng et al., 2025; Patil et al., 2025). Selective prediction evaluates risk under a learned reject option (Geifman & El-Yaniv, 2017; 2019). Our action set contains an explicit `HOLD` decision. Transport, syntax, schema, and semantic failures occupy a separate terminal category and remain errors end to end.

Research agents. ScienceAgentBench, CORE-Bench, and POPPER evaluate data analysis, reproducibility, and sequential falsification (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). Our research-evidence ontology covers one downstream state transition after evidence registration. Evidence discovery and end-to-end scientific work remain outside its scope.

3 METHOD

3.1 POLICY-DIFFERENCE COMPILATION

Let P be an executable policy, c a complete typed context, and $\pi_P(c)$ the prescribed action. For two policy versions and a finite grammar $\mathcal{X}$, define

$$\Delta(P_0, P_1; \mathcal{X}) = \{c \in \mathcal{X} : \pi_{P_0}(c) \neq \pi_{P_1}(c)\}.$$

Each $c \in \Delta$ yields a pair $((P_0, c), (P_1, c))$. The context, query, and output contract are identical; only the policy source changes. This construction makes policy-version sensitivity a necessary condition for pair correctness. A deterministic route invariant to the policy returns the same action twice, so at most one member can be correct and pair accuracy is zero.

The compiler associates every candidate with its witnessed changed rules and its directed action edge $(\pi_{P_0}(c), \pi_{P_1}(c))$. For suite T, adequacy is

$$A(T) = \left(\frac{|R(T)|}{|R_\Delta|}, \frac{|E(T)|}{|E_{\text{reach}}|} \right),$$

where R_Δ is the registered changed-rule inventory and E_{reach} is the action-edge set obtained by exhaustive execution over $\mathcal{X}$. Both certificates are relative to the published grammar.

Selecting the smallest suite with $A(T) = (1, 1)$ is a set-cover instance. We use a standard exact bit-mask dynamic program with $O(n2^{r+k})$ state updates for n candidates, r rules, and k edges. A specialized search selects one context per edge while tracking rule masks; a complete k-pair witness meets the edge lower bound and is globally minimum for the candidate pool. These solvers are checkable implementation components. Appendix A gives the proof and optimality certificate.

3.2 THREE SCORING ENDPOINTS

For scheduled item x, let $y(x) = \pi_P(c)$, let $e(x)$ be the action recovered by a strict parser, let $v(x)$ indicate full output-contract validity, and let $d(x)$ be the parsed structured decision. The parser accepts a parser-owned object or one strict top-level JSON object. It rejects repair, code-fence extraction, coercion, and duplicate keys. We report:

$$\begin{aligned} C_A(x) &= \mathbb{1}[e(x) = y(x)], && \text{strict action,} \\ C_C(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[e(x) = y(x)], && \text{contract action,} \\ C_F(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[d(x) = d^*(x)], && \text{full decision.} \end{aligned}$$

The task-specific contract contains five fields for Gatekeeper and six fields for the evidence-state tasks. Pair correctness is the product of the two member indicators for the selected endpoint. End-to-end metrics divide by every scheduled item or pair. Conditional action metrics divide by extractable or contract-valid items and complete pairs; they describe behavior after the indicated interface succeeds. Terminal failures are shown as `N` in confusion matrices, outside the policy action set. Response contents from failed cells are omitted.

4 EVIDENCE AND EVALUATION DESIGN

4.1 SOURCE-TRACEABLE REVISIONS

Table 1 separates policy revisions, verifier faults, authored stress instruments, and auxiliary robustness tests. The Gatekeeper study preserves exact old and new Rego source bytes, their parent and change commits, upstream regression-test names, and Apache-2.0 notices. OPA 1.19.1 reproduces both changes. We author a finite 12-context grammar around the upstream tests: six contexts for the FlexVolume empty-list change and six for the selector-kind change. Mapping zero violations to `ALLOW` and one or more violations to `HOLD` is an author policy mapping. The resulting 24 items were frozen before any route call. No human labels or agreement statistics enter the study.

The Trivy reproduction executes four upstream regression cases around the corrected inverted predicate. The Cosign reproduction exercises six cases and four verifier fault modes using the historical Go query logic and upstream tests. Cosign changes the verifier–policy interface and provides a distinct boundary-fault evidence class.

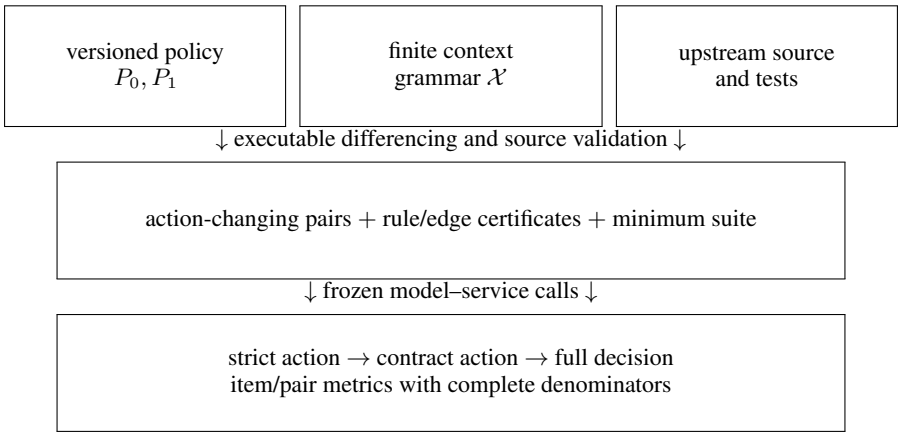


Figure 1: Benchmark and evidence flow. Source validation and suite construction finish before model calls. Each scoring layer answers a distinct deployment question.

Table 1: Evidence layers. Minimum sizes are exact relative to each committed candidate set.

Instrument	Source	Evidence role	Cases/pairs	Edges	Minimum
Gatekeeper	2 upstream Rego revisions	central policy revision	12 pairs	2/2	2
Trivy KCV0030	upstream predicate fix	policy-fault reproduction	4 cases	2/2	2
Cosign Rego verifier	upstream query fix	interface-fault reproduction	6 cases	2/2	4
Authored SLSA	12-rule policy migration	four-action stress test	60 pairs	10/10	10
Research r6	authored evidence contrasts	coverage addendum	150 pairs	–	–
Public challenge	preregistered evidence contrasts	robustness diagnostic	90 pairs	–	–

4.2 Route studies and auxiliary instruments

The central route study repeats all 24 Gatekeeper items three times on each of two hosted routes, for 144 scheduled calls. Each cell receives one request, zero client retries, and a 180-second timeout. The complete schedule, prompts, route labels, endpoint definitions, and no-retry rule were committed before the first call. The routes use different model, serving, and adapter stacks; their results are separate route measurements.

The authored SLSA instrument maps structured attestation fields to `ALLOW`/`HOLD`/`DEMOTE`/`ESCALATE` under two author-defined profiles (SLSA Steering Committee, 2026; in-toto Attestation Maintainers, 2026). Its rule-stratified suite covers 12 changed rules. Its graph suite covers every reachable directed action edge. Two pinned four-bit local routes provide a stress test of the compiled pairs. The deterministic executor supplies the labels and is the *definitional policy ceiling*. It is an oracle, while learned rows remain empirical controls.

The research-evidence ontology supplies two auxiliary suites. R6 contains 300 items with only eight `DEMOTE` and eight `ESCALATE` labels. The public challenge contains 90 pairs, 180 items, and 15 factors; it balances 90 `ALLOW` items against 30 each of `HOLD`, `DEMOTE`, and `ESCALATE`. Its cases and predictions were public before analysis, so it measures robustness on a public frozen challenge. A multinomial Naive Bayes sanity control is deliberately weak and provides no competitive baseline.

Table 2: Gatekeeper route results by repetition. All columns are end to end. A is strict action; F is exact full decision. Contract action equals A for every row.

Route	Repetition	A item	F item	A pair	F pair
Qwen	1	3/24	3/24	0/12	0/12
Qwen	2	2/24	2/24	0/12	0/12
Qwen	3	5/24	5/24	0/12	0/12
	total	10/72	10/72	0/36	0/36
DeepSeek Flash	1	15/24	15/24	6/12	6/12
DeepSeek Flash	2	19/24	18/24	7/12	6/12
DeepSeek Flash	3	19/24	19/24	8/12	8/12
	total	53/72	52/72	21/36	20/36

5 RESULTS

5.1 SOURCE VALIDATION AND COMPILED ADEQUACY

OPA execution reproduces both Gatekeeper behavior changes on all 12 contexts. The suite covers 2/2 changed rules and both `ALLOW`→`HOLD` and `HOLD`→`ALLOW` transitions. One pair realizes one edge, and a two-pair witness also covers both rules; the witness meets the two-edge lower bound. The Trivy cases reproduce two transitions in each direction around the inverted predicate, with a two-case minimum. The Cosign cases reproduce four documented verifier fault modes and both directions; four cases are required to cover the four modes. These source results establish concrete fault instances and exact finite-grammar coverage.

The authored four-action migration exposes a separate adequacy result. Its 12-rule stratified suite covers 12/12 changed rules and 6/10 reachable edges. The graph suite covers 12/12 rules and 10/10 edges. An exact search selects a 10-pair, 20-call joint suite, reducing the 120-call graph suite by 83.3%. The two local routes score 10/60 and 2/60 pairs on the full graph; both fail the 20-call minimum gate. This study validates the compiler on a richer action graph while retaining authored policy scope.

5.2 REPEATED ROUTE BEHAVIOR ON UPSTREAM REVISIONS

Table 2 reports every scheduled Gatekeeper cell. Strict and contract-action scores coincide because every strictly extracted object also satisfies the task contract. The DeepSeek route records 15/24, 19/24, and 19/24 strict actions and totals 53/72 items and 21/36 pairs; 52/72 decisions and 20/36 pairs are fully exact. The Qwen route totals 10/72 strict and full decisions and completes no pair. Every extracted action on both routes matches the OPA action. Conditional strict-action item and pair accuracy is therefore 1.0 whenever a complete pair exists.

The terminal failures identify the limiting interface behavior without exposing response content. The Qwen route produces 62 Markdown-fenced objects, all rejected by the preregistered strict parser. The DeepSeek route records 12 read timeouts, three fenced objects, three non-JSON prefixes or envelopes, and one length-terminated non-JSON response. The protocol performs zero response repair or rescoring. Across repeated cells, 13/24 DeepSeek items are strict-action correct in all three repetitions, compared with 0/24 for Qwen; 5/12 DeepSeek pairs pass all three repetitions.

5.3 ENDPOINT SENSITIVITY AND EVIDENCE-STATE DIAGNOSTICS

The offline dual-endpoint audit covers 1,320 previously scheduled item cells across ten route stacks. Strict extraction identifies 69 actions outside the full six-field contract. Of these, 49 are correct and create 40 additional correct pairs. The largest gap occurs on the fixed SLSA Qwen route: strict action yields 120/120 items and 60/60 pairs, while contract action yields 88/120 and 28/60. The balanced Flash route changes from 105/180 items and 19/90 pairs under strict action to 90/180 and 13/90 under the full contract. These endpoints answer different integration questions; Table 3 reports both where records permit offline extraction.

Table 3: Auxiliary end-to-end (E2E) and contract-conditional metrics. "–" marks an unavailable post-hoc strict-action audit for r6.

Route	Valid	Strict item	Strict pair	Contract item	Contract pair	Cond. item/pair
r6 Qwen	272/300	–	–	258/300	113/150	258/272; 113/123
Challenge Qwen	148/180	132/180	48/90	132/180	48/90	132/148; 48/64
Challenge Flash	146/180	105/180	19/90	90/180	13/90	90/146; 13/61
SLSA Qwen	88/120	120/120	60/60	88/120	28/60	88/88; 28/28
SLSA Flash	108/120	110/120	51/60	108/120	49/60	108/108; 49/49

Table 4: End-to-end contract-action recall by gold action.

Route	ALLOW	HOLD	DEMOTE	ESCALATE
r6 Qwen	143/150	110/134	0/8	5/8
Challenge Qwen	79/90	26/30	4/30	23/30
Challenge Flash	68/90	9/30	0/30	13/30

Action support changes the diagnosis. Table 4 gives end-to-end contract-action recall. R6 Qwen has only eight examples for each rare action and recovers 0/8 DEMOTE and 5/8 ESCALATE. The balanced challenge increases each non-ALLOW action to 30 cases: Qwen recovers 4/30 DEMOTE, and Flash recovers 0/30. The complete confusion matrices in Appendix C retain terminal failures in every row.

The historical r6 Flash lane used an obsolete frozen service interface. All 300 requests returned HTTP 401 before model output, giving 0/300 coverage, 0/300 end-to-end items, and 0/150 end-to-end pairs; conditional metrics are undefined. The later Flash routes use a different service and output stack. Their successful calls establish executability for those later paths and carry no causal comparison with the 401 lane.

6 Discussion

Executable revisions provide focused regression oracles. The upstream cases show that small source edits can reverse admission outcomes for fixed inputs. The compiler turns those reversals into tests that a policy-invariant route cannot pass in pairs. Changed-rule and action-edge certificates reveal distinct omissions, and the minimum suite offers a low-cost CI gate before a fuller diagnostic run. This contribution is an integration of semantic policy differencing with model-service evaluation; Cirg supplies the closest change-impact precedent, while GuideBench and TDAD cover guideline following and evolving behavioral specifications.

Interface reliability is part of observed route behavior. All valid Gatekeeper actions are correct, yet the two routes finish only 21/36 and 0/36 pairs under strict extraction. The failure taxonomy attributes terminal losses to service delay and output envelopes; every extracted action is correct. A downstream system that needs only an action may choose the strict-action endpoint. A system that consumes the complete object must use contract or full-decision scores. The 1,320-item audit shows that collapsing these endpoints can change pair results substantially.

Balanced actions expose failures hidden by aggregate accuracy. The public challenge keeps 180 calls in the denominator and raises each non-ALLOW action to 30. Its 4/30 and 0/30 DEMOTE recalls qualify the apparently stronger aggregate scores. R6 remains a coverage addendum with 8/300 support for each rare action. Public preregistration makes the challenge a robustness check under known cases; it supplies no hidden-distribution estimate.

The intended deployment boundary is narrow. The benchmark targets existing model-mediated structured decisions during policy maintenance. A direct deterministic executor remains the preferred enforcement path for fully structured inputs. The definitional policy ceiling records the task specification. The weak Naive Bayes row checks surface shortcuts and provides only a sanity control.

7 LIMITATIONS

The model-service study covers two Gatekeeper revisions, one author-built context grammar, two hosted routes, and three repetitions. Trivy and Cosign add source validity without route replication. The authored SLSA migration broadens the action graph while preserving synthetic policy scope. Together these instruments demonstrate a testing construction and concrete failure instances. Population estimates for policies, model families, and deployments remain open.

The Gatekeeper mapping from Rego violations to `ALLOW`/`HOLD` and every SLSA or research-evidence label are author policy mappings. No human annotation study was conducted. A review of 12 source-traceable naturalistic candidates yielded zero admissible cases because required time-stamped artifacts and provenance were absent; freezing produced zero model calls. Naturalistic validity therefore remains open.

The finite grammar bounds every coverage claim. An omitted context can hide additional behavior, and incorrect rule witnesses can invalidate a rule-coverage certificate. The Trivy reproduction uses a documented compatibility helper to execute historical Rego under the current runtime while preserving upstream policy bytes. The Cosign study targets verifier logic, a separate evidence class from policy-source differencing.

Hosted route results combine model weights, prompts, provider adapters, output constraints, service load, and time. The three repetitions describe this fixed schedule and provide no independent model-family sample. Between-route differences are descriptive. The old Flash 401 lane records an obsolete interface failure; later success speaks only to the later path.

The balanced challenge is public and shares an authored ontology with r6. Its Qwen and Flash pair scores are 48/64 and 13/61 conditional versus 48/90 and 13/90 end to end. The r6 Qwen addendum gives 113/123 conditional pairs versus 113/150 end to end. Invalid contents are excluded from the manuscript, while their terminal classes remain in denominators and receipts. Canonical ledgers also leave policy retrieval, ambiguous source text, evidence extraction, and downstream utility outside the evaluation.

8 CONCLUSION

CLAIMPROMOTEBENCH compiles executable policy revisions into paired model-service regression tests with rule, transition, and minimum-suite certificates. Two real Gatekeeper revisions yield a two-pair minimum and expose large route-level completion gaps across three repetitions. A 1,320-item endpoint audit shows that strict action and full-contract scoring can diverge. R6 Qwen reaches 113/150 pairs end to end versus 113/123 conditional; balanced Qwen and Flash reach 48/90 and 13/90 versus 48/64 and 13/61, with `DEMOTE` recall of 4/30 and 0/30. The resulting evidence supports CI testing of finite authored policy surfaces and structured service interfaces.

9 REPRODUCIBILITY STATEMENT

The artifact binds released policies, datasets, prompts, upstream source bytes, commit identifiers, licenses, finite grammars, frozen schedules, and route configurations. It also binds terminal outcome classes, analyses, and privacy receipts. Validators recompute coverage, minimum suites, endpoint metrics, action recalls, and confusion matrices from released records. The anonymous derivative excludes raw failed responses, free-form errors, provider response identifiers, hidden reasoning, account identity, and private commit identity. Its manifest derives file counts from the generated tree. Completed one-shot studies remain frozen.

10 AI ASSISTANCE DISCLOSURE

Generative AI tools assisted literature search, planning, code and benchmark drafting, debugging, analysis checks, and manuscript editing. The author chose the research question, policies, protocols, exclusions, analyses, and claims; verified the datasets, results, artifacts, and references; and is responsible for the manuscript. AI systems are neither authors nor human annotators.

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A Compiler Guarantees

Minimum-suite hardness. Take a Set Cover instance with universe U and subsets $S_1, \ldots, S_n$. Create one required rule for each element of U and one candidate context for each S_i whose rule-witness set is S_i. Assign every candidate the same required action edge. Any nonempty selection covers that edge, and it covers all required rules exactly when the corresponding subsets cover U. A minimum joint suite therefore gives a minimum set cover.

Exact dynamic program. Index the $r + k$ required rules and edges and encode coverage as a bit mask. After processing the first j candidates, retain for every reachable mask the shortest subset, breaking ties by context identifier. Excluding candidate $j + 1$ preserves each state; including it unions its coverage mask with each prior mask. Induction on j proves that the final full-mask representative is cardinality-minimum. The implementation matches exhaustive enumeration on 64 deterministic small candidate pools.

Edge lower bound. Each pair realizes one directed edge. Covering k edges requires at least k pairs. If the specialized search selects one candidate per edge and also covers every rule, its k-pair witness meets the lower bound. This certifies the two-pair Gatekeeper suite and the ten-pair authored SLSA suite relative to their committed candidate pools.

B Failure Taxonomy and Scope Notes

The repeated Gatekeeper study classifies failures from terminal protocol metadata only. Qwen has 62 `MARKDOWN_CODE_FENCE` failures. DeepSeek Flash has 12 `READ_TIMEOUT`, three `MARKDOWN_CODE_FENCE`, three `NON_JSON_PREFIX_OR_ENVELOPE`, and one `LENGTH_TERMINATED_NON_JSON` failure. The diagnostic loads no gold labels, raw response text, hidden reasoning, free-form error strings, or provider response identifiers.

The naturalistic-candidate review applied source-traceability criteria before constructing any prompt. Twelve candidates lacked the complete time-stamped artifacts and provenance required for the author policy mapping. The frozen outcome is zero eligible cases and zero model calls; the artifact claims no annotators and no agreement statistic.

C Complete Action-Level Results

Table 5: End-to-end action recall for every route summarized in the main text. "–" marks actions outside a task's label set.

Configuration and endpoint	ALLOW	HOLD	DEMOTE	ESCALATE
Gatekeeper Qwen, strict	4/36	6/36	–	–
Gatekeeper Flash, strict	32/36	21/36	–	–
r6 Qwen, contract	143/150	110/134	0/8	5/8
Challenge Qwen, strict	79/90	26/30	4/30	23/30
Challenge Qwen, contract	79/90	26/30	4/30	23/30
Challenge Flash, strict	68/90	10/30	2/30	25/30
Challenge Flash, contract	68/90	9/30	0/30	13/30
SLSA Qwen, strict	60/60	20/20	20/20	20/20
SLSA Qwen, contract	60/60	20/20	1/20	7/20
SLSA Flash, strict	59/60	17/20	17/20	17/20
SLSA Flash, contract	59/60	17/20	15/20	17/20
Graph Mistral, contract	9/36	18/24	0/24	13/36
Graph Phi-4, contract	21/36	5/24	0/24	1/36

Table 6: Gatekeeper aggregate strict-action confusion counts over three repetitions (left) and authored SLSA transition-graph contract-action confusion counts (right). N marks the endpoint's terminal failure category.

Gatekeeper

Route	Gold	ALLOW	HOLD	N
Qwen	ALLOW	4	0	32
Qwen	HOLD	0	6	30
Flash	ALLOW	32	0	4
Flash	HOLD	0	21	15

Authored transition graph

Route	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Mistral	ALLOW	9	8	0	9	10
Mistral	HOLD	2	18	0	4	0
Mistral	DEMOTE	3	0	0	0	21
Mistral	ESCALATE	2	3	0	13	18
Phi-4	ALLOW	21	0	0	0	15
Phi-4	HOLD	1	5	0	0	18
Phi-4	DEMOTE	3	12	0	0	9
Phi-4	ESCALATE	0	0	0	1	35

Table 7: R6 and balanced-challenge full-contract confusion counts. Columns are terminal predicted actions; N is a contract failure.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
r6 Qwen	ALLOW	143	0	0	0	7
r6 Qwen	HOLD	4	110	0	2	18
r6 Qwen	DEMOTE	8	0	0	0	0
r6 Qwen	ESCALATE	0	0	0	5	3
Challenge Qwen	ALLOW	79	0	0	0	11
Challenge Qwen	HOLD	0	26	0	0	4
Challenge Qwen	DEMOTE	8	4	4	4	10
Challenge Qwen	ESCALATE	0	0	0	23	7
Challenge Flash	ALLOW	68	4	0	8	10
Challenge Flash	HOLD	13	9	0	0	8
Challenge Flash	DEMOTE	25	3	0	0	2
Challenge Flash	ESCALATE	0	3	0	13	14

Table 8: Balanced-challenge strict-action confusion counts. N means no safely extracted action.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Challenge Qwen	ALLOW	79	0	0	0	11
Challenge Qwen	HOLD	0	26	1	0	3
Challenge Qwen	DEMOTE	8	4	4	4	10
Challenge Qwen	ESCALATE	0	0	0	23	7
Challenge Flash	ALLOW	68	4	5	13	0
Challenge Flash	HOLD	13	10	7	0	0
Challenge Flash	DEMOTE	25	3	2	0	0
Challenge Flash	ESCALATE	0	3	2	25	0

Table 9: Fixed SLSA strict-action and contract-action confusion counts. N is the endpoint's terminal failure category.

Route	Endpoint	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Qwen	strict	ALLOW	60	0	0	0	0
Qwen	strict	HOLD	0	20	0	0	0
Qwen	strict	DEMOTE	0	0	20	0	0
Qwen	strict	ESCALATE	0	0	0	20	0
Qwen	contract	ALLOW	60	0	0	0	0
Qwen	contract	HOLD	0	20	0	0	0
Qwen	contract	DEMOTE	0	0	1	0	19
Qwen	contract	ESCALATE	0	0	0	7	13
Flash	strict	ALLOW	59	0	0	0	1
Flash	strict	HOLD	0	17	0	0	3
Flash	strict	DEMOTE	0	0	17	0	3
Flash	strict	ESCALATE	0	0	0	17	3
Flash	contract	ALLOW	59	0	0	0	1
Flash	contract	HOLD	0	17	0	0	3
Flash	contract	DEMOTE	0	0	15	0	5
Flash	contract	ESCALATE	0	0	0	17	3